

Name: _____

Date: _____

Resolving Vectors

SYNOPSIS

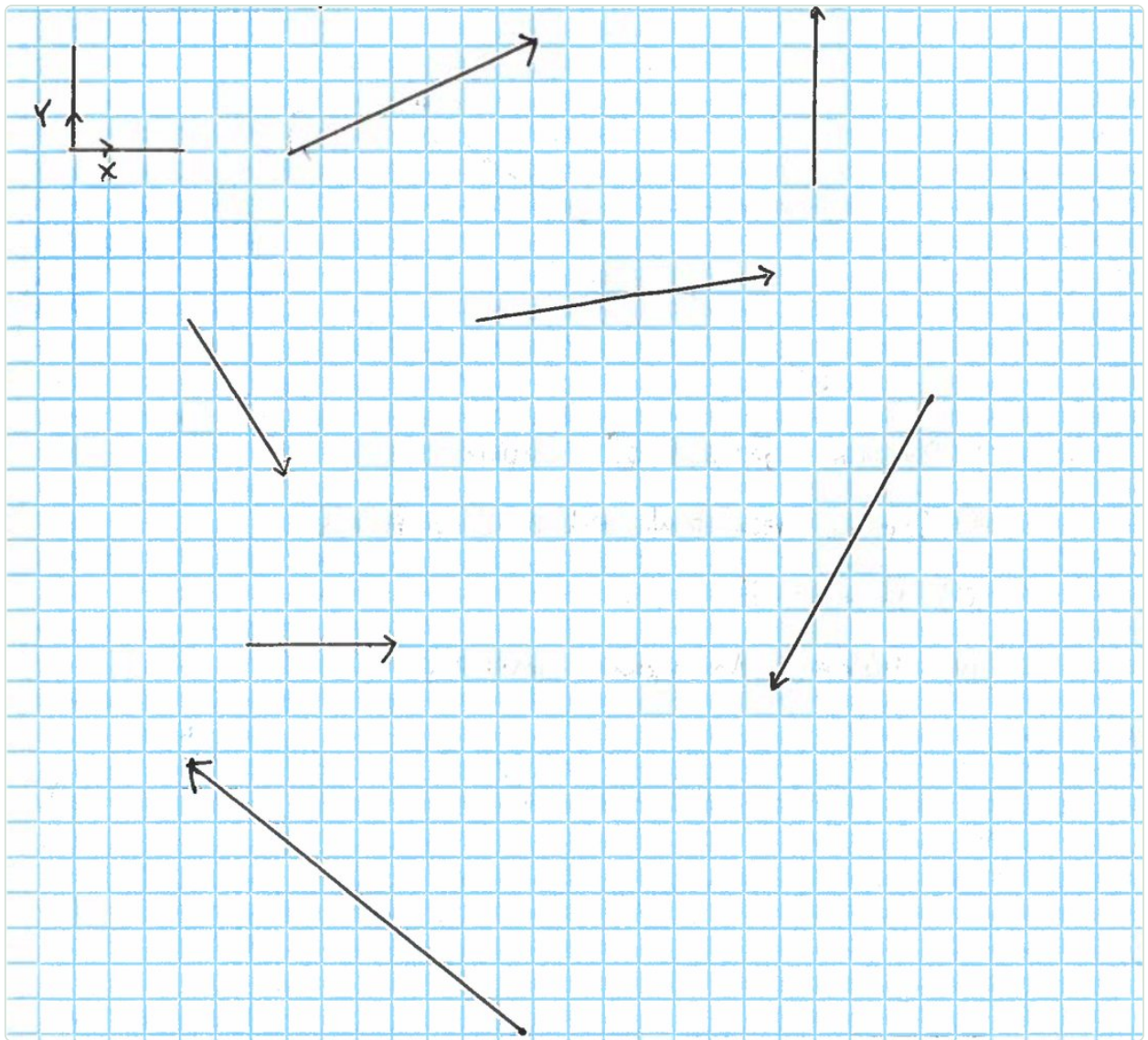
We can add vectors using the vector addition rule (the head-to-tail rule). Going in reverse, we can *resolve*, or split up, a vector into *horizontal* and *vertical* components. In this note, we practice resolving vectors.

PERFORMANCE EXPECTATIONS

Resolve the vectors by drawing the horizontal and vertical components for given vectors.

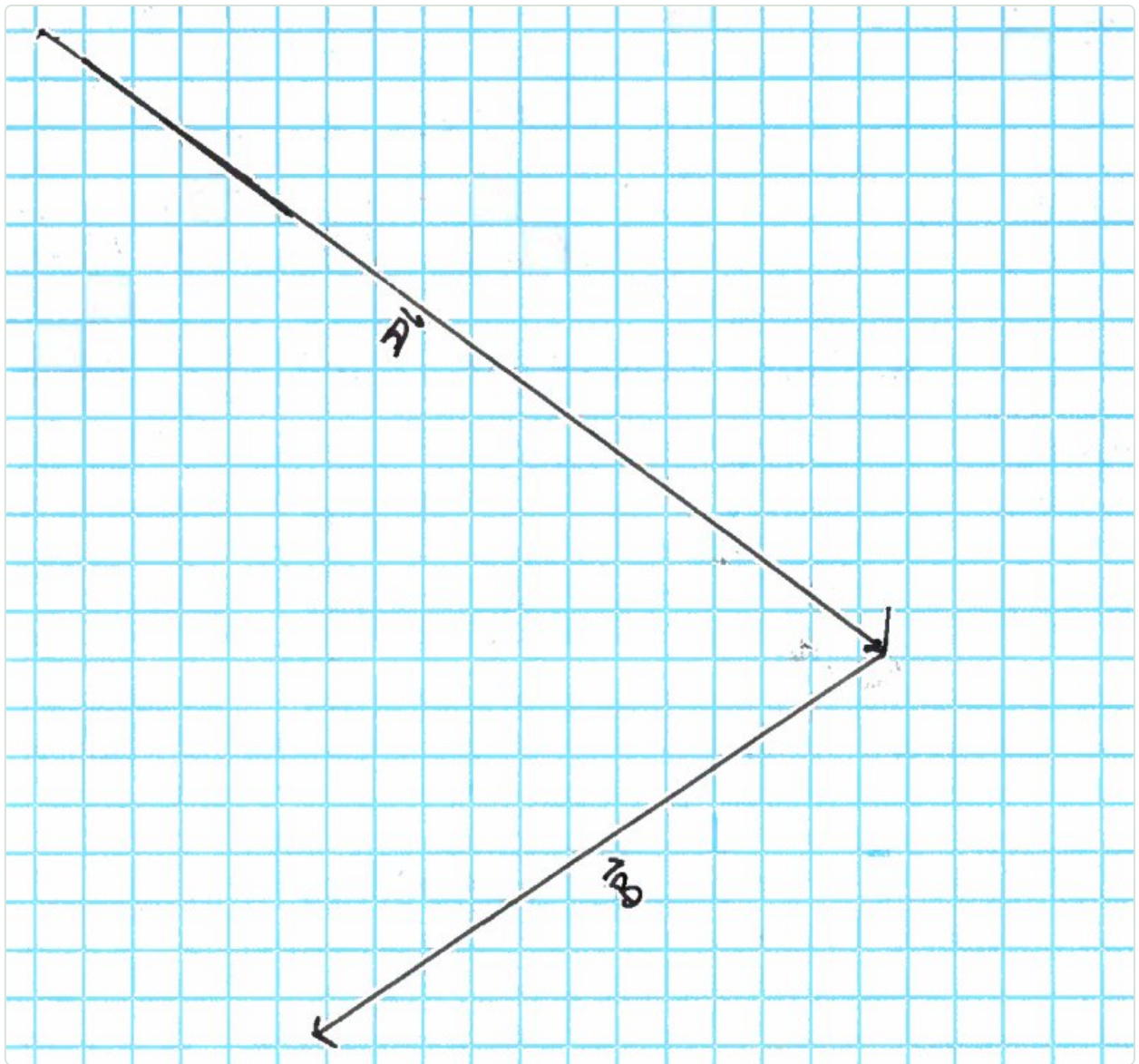
Instructions and Questions

1. Use a ruler to draw the horizontal (along x) and vertical (along y) components.



Draw the horizontal and vertical component of each vector above.

2. Add vectors A and B to find the vector C. C is the resultant vector.
3. Resolve vectors A and B.
4. Resolve the resultant vector C.
5. Answer the questions below.



Add $A + B$ to find C , then resolve A , B , and C into components.

Q1) Add ALL horizontal components of the vectors A and B and compare the sum of all horizontal components to the horizontal component of C . What do you notice?

Q2) Add ALL vertical components of the vectors A and B and compare the sum of all vertical components to the vertical component of C . What do you notice?

Q3) What can you say about the **sum** of the components and the components of the resultant vector?
